

Home Assignment for Software Analyst Position

Overview

Create a Python application that manages Notepad++ installations on Windows systems. The application should check the current installed version, compare it with the latest available version, and update it if necessary. No graphical user interface is required - this should be implemented as a command-line application.

Technical Requirements

1. Code Quality

- Implement the solution using Python 3.8+
- Organize code into logical modules/packages
- Include clear documentation (docstrings, comments, etc.)

2. Static Analysis

- Code must pass validation with a linter of your choice (flake8, pylint, etc.)
- Code must pass validation with a static type checker of your choice (mypy, pyright, etc.)
- Configuration files for these tools should be included in the project

3. Testing

- Implement comprehensive unit tests
- Include test cases for both success and error scenarios
- Tests should be runnable with a single command

Deliverables

1. Complete Python project with all source code
2. README.md with:
 - Project overview
 - Installation instructions
 - Usage instructions
 - Explanation of design decisions
 - Description of testing approach
3. Requirements.txt or pyproject.toml for dependencies
4. Test suite with instructions to run tests
5. Configuration files for linting and type checking

Submission Guidelines

- Submit your solution as a GitHub repository or as an archive file
- Ensure the repository is public or provide access to the reviewers
- Include a brief summary of your approach and any challenges faced